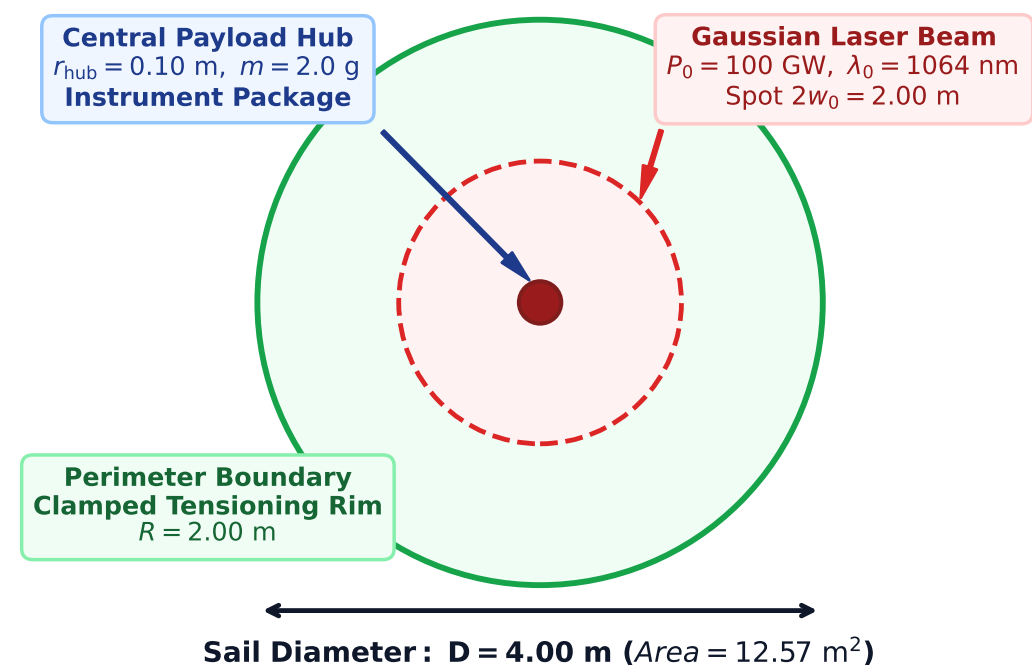
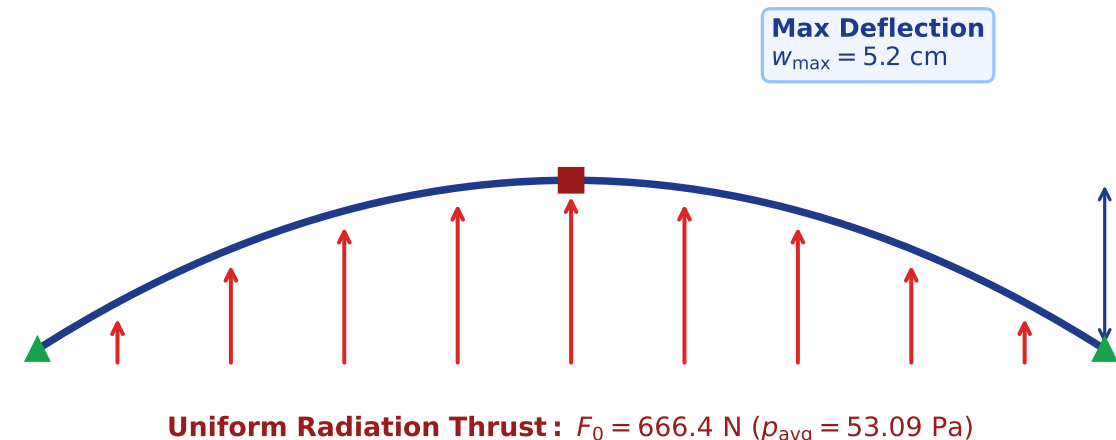


A. Macroscale Top-Down Blueprint



B. Side Elevation & Spherical Catenary Profile



C. Nanoscale 12-Bilayer Metasurface Stack

Quarter-Wave Bragg Layer Design :

- High-Index Layer (TiO_2) : $n_H = 2.48$, $d_H = \frac{\lambda_0}{4n_H} = 107.25 \text{ nm}$
- Low-Index Layer (SiO_2) : $n_L = 1.45$, $d_L = \frac{\lambda_0}{4n_L} = 183.45 \text{ nm}$
- Bilayer Period : $\Lambda = d_H + d_L = 290.70 \text{ nm}$ | Coating : $t_{\text{stack}} = 3.49 \mu\text{m}$

